

2 AI Fundamentals Quiz

1. Which definition of AI was adopted in the presentation?

- A) Computer systems performing tasks typically associated with intelligence such as problem solving, decision making, and forecasting
- B) Any electronic spreadsheet
- C) Only humanoid robots
- D) A database with no predictive capability

ANS:

2. Which system is recognized in the slides as the first chatbot in modern AI history?

- A) Perceptron
- B) ELIZA
- C) Expert system
- D) Transformer

ANS:

3. In the historical timeline, which term describes the period when AI funding and interest dropped significantly?

- A) Symbolic AI
- B) Deep learning boom
- C) AI winter
- D) Expert systems era

ANS:

4. Which task predicts categories or “classes” from input features?

- A) Regression
- B) Classification
- C) Compression
- D) Clustering only

ANS:

5. Which task predicts numerical outputs from input features?

- A) Classification
- B) Tokenization
- C) Regression
- D) Authentication

ANS:

6. Which task focuses on identifying outliers or abnormal data points?

- A) Sequence modeling
- B) Anomaly detection
- C) Supervised learning
- D) Deep learning

ANS:

7. What does generative modeling do according to the slides?

- A) Deletes all training examples
- B) Predicts only binary labels
- C) Prevents new samples
- D) Creates new data that resembles the training data

ANS:

8. Which sequence best describes the typical machine-learning pipeline shown in the presentation?

- A) Collect data, label, choose architecture, train, evaluate, deploy
- B) Deploy, then define the task

- C) Evaluate before collecting data
- D) Remove data and skip training

ANS:

9. What type of data does supervised learning use?

- A) Only unlabeled examples
- B) Only agent-generated rewards
- C) Labeled examples
- D) No data

ANS:

10. What is the main goal of unsupervised learning in the slides?

- A) Follow explicit output labels
- B) Discover patterns and relationships in unlabeled data
- C) Maximize a reward through actions
- D) Compile source code

ANS:

11. How does reinforcement learning obtain its learning signals?

- A) Only from static labels
- B) Only from clustering
- C) From an agent's interaction with an environment and the resulting rewards
- D) From manual rules alone

ANS:

12. What core model structure does deep learning use according to the presentation?

- A) Hand-written rules only
- B) Decision trees only
- C) Neural networks composed of layers of interconnected nodes
- D) Fixed lookup tables

ANS:

13. Which advantage of deep learning is highlighted as "performance" in the slides?

- A) Guarantee of zero error in all tasks
- B) Superhuman performance in some domains such as image recognition, chess, and Go
- C) Requirement for no data
- D) Perfect transparency in all models

ANS:

14. Which point illustrates the "real-world usefulness" of deep learning in the slides?

- A) It is only used in board games
- B) It has a wide variety of useful applications, including healthcare, social media, and autonomous vehicles
- C) It is limited to symbolic logic
- D) It cannot be deployed

ANS:

15. Which statement captures the "scalability" property of deep learning mentioned in the presentation?

- A) Performance usually gets worse with more data
- B) Performance quickly saturates and cannot improve
- C) Deep-learning models are highly scalable and performance often improves with more data rather than tapering off
- D) Deep-learning models cannot use large datasets

ANS: